

AI CAREER ASSISTANT TOOL

PROJECT REPORT

ABSTRACT

The AI Career Assistant Tool is a web-based application developed using Python and Flask that helps students and job seekers evaluate their resumes and improve their chances of securing employment. The system analyses resumes uploaded in PDF, DOCX, and TXT formats, extracts relevant information, identifies technical skills, compares them with job descriptions, calculates ATS (Applicant Tracking System) scores, recommends suitable job roles, and generates personalized learning roadmaps.

The project aims to simplify resume screening and provide career guidance through automation. It assists users in understanding skill gaps, improving resume quality, and preparing for interviews. The application provides a user-friendly interface and generates valuable insights that help users align their profiles with industry requirements.

CHAPTER 1: INTRODUCTION

1.1 Overview

- The recruitment process has become highly competitive, and organizations increasingly use Applicant Tracking Systems (ATS) to filter resumes before they reach human recruiters. Many qualified candidates fail to pass ATS screening because their resumes do not contain the required keywords or skills.
- The AI Career Assistant Tool is designed to help job seekers understand how well their resumes match industry requirements. It provides ATS scoring, job description matching, skill analysis, career recommendations, and interview preparation guidance.

1.2 Need for the Project

- Many students and fresh graduates face difficulties in creating industry-ready resumes. They often lack awareness about the skills required for specific job roles.

This project addresses these challenges by:

- Evaluating resumes automatically
- Identifying missing skills
- Providing ATS compatibility scores
- Recommending suitable job opportunities
- Suggesting learning roadmaps

1.3 Problem Statement

Job seekers often struggle to determine whether their resumes meet recruiter expectations. Manual resume evaluation is time-consuming and subjective. There is a need for an intelligent system that can automatically analyze resumes and provide career guidance.

1.4 Objectives

The objectives of this project are:

- Resume parsing and analysis
- ATS score calculation
- Job description matching
- Skill extraction
- Missing skill identification
- Job recommendation generation
- Learning roadmap creation
- Interview question suggestion

1.5 Scope of the Project

The project can be used by:

- Students
- Fresh graduates
- Job seekers
- Career counselors
- Training institutions

CHAPTER 2: SYSTEM ANALYSIS

2.1 Existing System

Traditional resume evaluation methods involve manual review by recruiters.

Limitations

- Time consuming
- Human errors
- Inconsistent evaluation
- Lack of personalized guidance
- No ATS compatibility checking

2.2 Proposed System

The AI Career Assistant Tool automates resume analysis using predefined skill matching and ATS scoring mechanisms.

Features

- Resume upload
- PDF/DOCX/TXT support
- ATS score generation

- JD matching
- Skill extraction
- Missing skill analysis
- Job recommendations
- Learning roadmap generation

2.3 Advantages

- Faster resume evaluation
- Reduced manual effort
- Improved ATS compatibility
- Personalized recommendations
- Better career guidance

• CHAPTER 3: SYSTEM DESIGN

3.1 System Architecture

- Resume Upload
- Text Extraction
- Skill Detection
- Job Description Analysis
- ATS Score Calculation
- Missing Skill Detection
- Job Recommendation
- Learning Roadmap
- Dashboard Display

3.2 Flowchart

- Start
- Upload Resume
- Extract Text
- Analyze Skills
- Upload/Paste JD
- Compare Skills
- Calculate ATS Score
- Generate Recommendations
- Display Results
- End

CHAPTER 4: SOFTWARE REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software Requirements

- Windows 10/11
- Python 3.10+
- Flask
- Visual Studio Code
- GitHub

CHAPTER 5: TOOLS AND TECHNOLOGIES

Python

Python is the primary programming language used for backend development.

Flask

Flask is a lightweight web framework used to create the web application.

HTML

HTML is used to design web pages.

CSS

CSS is used for styling the application.

Bootstrap

Bootstrap provides responsive UI components.

PyPDF2

Used for extracting text from PDF files.

Python-Docx

Used for extracting text from DOCX files.

GitHub

Used for source code management and version control.

CHAPTER 6: MODULE DESCRIPTION

6.1 Resume Upload Module

Allows users to upload resumes.

Supported formats:

- PDF
- DOCX
- TXT

6.2 Resume Parsing Module

Extracts text content from uploaded resumes.

Functions:

- `extract_pdf()`
- `extract_docx()`
- `extract_txt()`

6.3 Skill Detection Module

Compares resume content with predefined skills.

Skills Database:

- Python
- Java
- SQL
- HTML
- CSS
- JavaScript
- Flask
- Django
- Git

- AWS
- Docker
- Machine Learning

6.4 ATS Score Module

Calculates ATS score based on:

- Skills found
- Projects
- Internship experience
- Certifications

6.5 Job Matching Module

Compares resume skills with job description requirements.

6.6 Recommendation Module

Generates recommended job roles.

Examples:

- Python Developer
- Data Analyst
- AI Engineer

6.7 Learning Roadmap Module

Suggests skills to learn.

Example:

- Week 1 → Learn Flask
- Week 2 → Learn Django

6.8 Interview Question Module

Provides technical and HR interview question

CHAPTER 7: IMPLEMENTATION

Backend Development

- The backend was developed using Flask.

Resume Upload

- `resume = request.files["resume"]`

PDF Extraction

- `def extract_pdf(path):`

DOCX Extraction

- `def extract_docx(path):`

Skill Matching

- `skills_db = [...]`

ATS Score Calculation

- `score += min(len(found_skills) * 5, 50)`

Job Recommendation

- `recommended_jobs = []`

Learning Roadmap

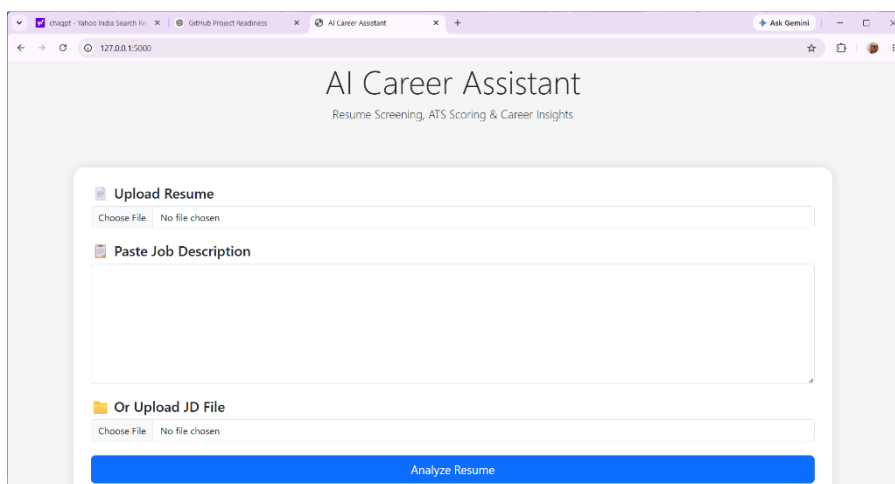
- `roadmap.append(...)`
-

CHAPTER 8: USER INTERFACE

Home Page

Features:

- Upload Resume
- Paste Job Description
- Upload JD File
- Analyse Resume Button



Home Page Screenshot

Description:

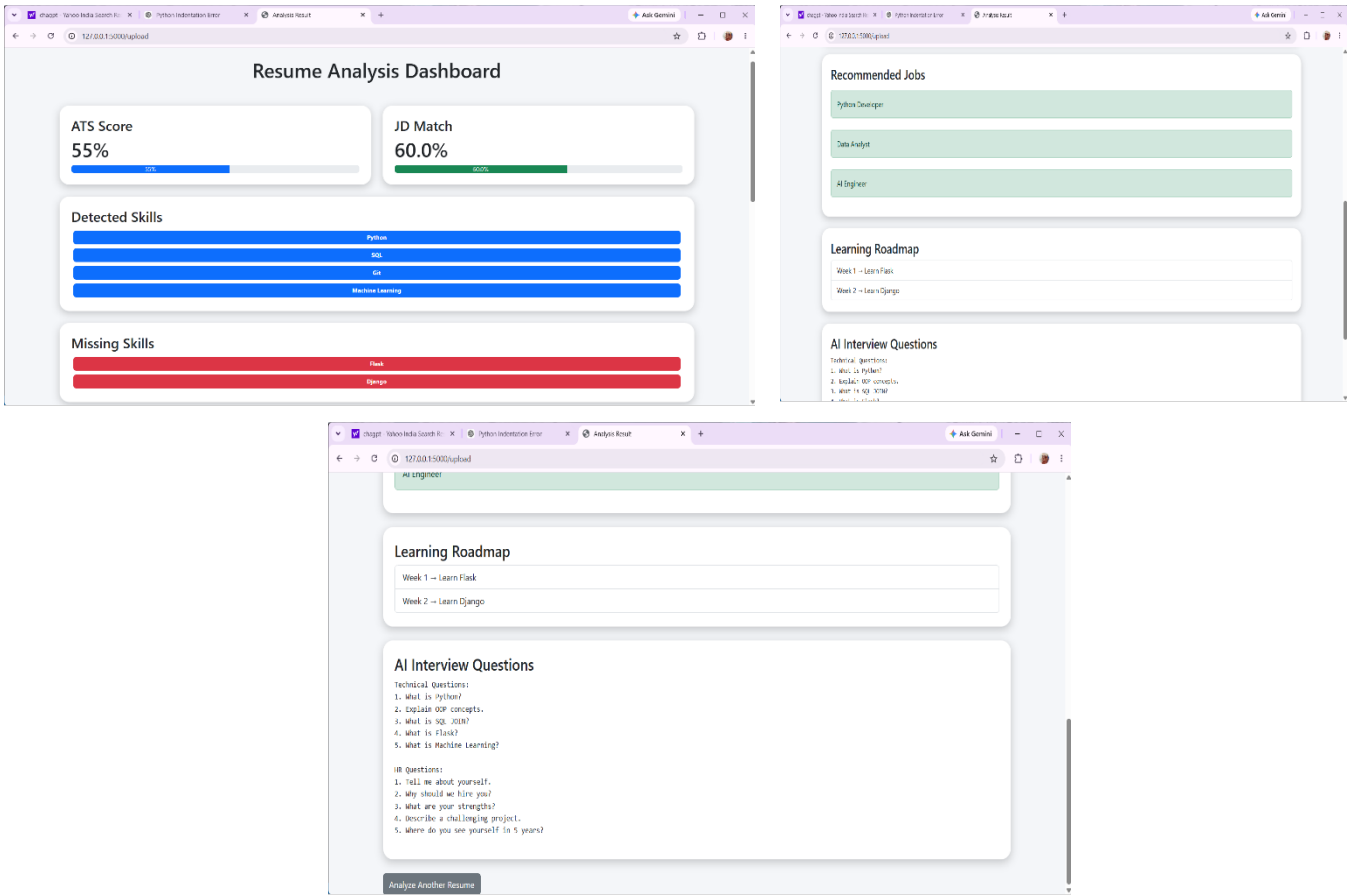
The home page provides options to upload resumes and job descriptions for analysis.

Result Dashboard

Features:

- ATS Score
- JD Match Percentage
- Detected Skills
- Missing Skills
- Recommended Jobs

- Learning Roadmap
- Interview Questions



(Result Dashboard Screenshot)

Description:

The dashboard displays detailed analysis and recommendations.

CHAPTER 9: TESTING

Iteration 1

1.Date & Time: 01.06.2026 & 4:00PM

2.Issue

Problems occurred during PDF text extraction.

3.Solution

Implemented improved extraction logic using PyPDF2.

4.Result

Successful resume parsing.

Iteration 2

1.Date & Time: 05.06.2026 & 12:00PM

2.Issue

ATS scores were inconsistent.

3.Solution

Improved skill matching and scoring logic.

4.Result

More accurate ATS score generation.

CHAPTER 10: RESULTS

Sample Output:

ATS Score

55%

JD Match

60%

Skills Detected

- Python
- SQL
- Git
- Machine Learning

Missing Skills

- Flask
- Django

Recommended Jobs

- Python Developer
- Data Analyst
- AI Engineer

Learning Roadmap

- Week 1 → Learn Flask
- Week 2 → Learn Django

CHAPTER 11: ADVANTAGES

- Easy to use
- Automated analysis
- Saves recruiter time
- Improves ATS compatibility
- Provides career guidance
- Identifies skill gaps
- Supports multiple resume formats

CHAPTER 12: FUTURE ENHANCEMENTS

- AI Resume Builder
- Resume Rewrite Suggestions
- LinkedIn Integration
- Real-Time Job APIs
- AI Career Chatbot
- Interview Simulator
- Multi-language Support

CHAPTER 13: CONCLUSION

- The AI Career Assistant Tool was successfully designed and implemented using Python and Flask. The system automates resume analysis, ATS scoring, job description matching, and career guidance.
- It provides users with valuable insights regarding their strengths and weaknesses while helping them improve their employability.
- The project demonstrates the practical application of Artificial Intelligence concepts in career development and recruitment assistance.
- The developed system serves as a useful tool for students, graduates, and job seekers seeking to enhance their resumes and prepare for future job opportunities.